

Subtraction and Negative Signs: Diagnosing the Error

Answer Key

Grade 6–7 The Number System

Quick Answers

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|-------|----------|--------------|----------|
| 1. -5 | 3. -5.75 | 5. -9, -5, 5 | 7. -11 |
| 2. -7 | 4. — | 6. 5 | 8. -0.75 |
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Common wrong answers

- If they got -11: read the two negatives as one and computed $-8 + (-3)$*
 - If they got 7: subtracted the smaller number from the larger one and kept the answer positive*
 - If they got -1.25: changed the subtraction to addition without changing the sign of the second number*
 - If they got -13: added the two negative numbers instead of subtracting a negative*
 - If they got -3: turned the subtraction into an addition of a positive, computing $-7 + 4$*
 - If they got -12.75: dropped the double negative and subtracted 6 instead of adding it*
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1. Evaluate $-8 - (-3)$.

Solution: Add the opposite. The second number is -3 , and its opposite is $+3$, so $-8 - (-3) = -8 + 3$. Starting at -8 and moving 3 units to the right lands on -5 . -5

Watch for: an answer of -11 means the two negatives were merged into one and the student computed $-8 + (-3)$. The minus sign in front of the parenthesis is an operation; the minus inside is part of the number.

2. Evaluate $5 - 12$.

Solution: Add the opposite: $5 - 12 = 5 + (-12)$. The two numbers pull in opposite directions, so subtract distances from zero, $12 - 5 = 7$, and take the sign of the number farther from zero, which is -12 . The result is therefore negative. -7

Watch for: an answer of 7 means the student computed $12 - 5$ to avoid the negative. Which number is written first is what decides the sign.

3. Evaluate $-3.5 - 2.25$.

Solution: Add the opposite: $-3.5 - 2.25 = -3.5 + (-2.25)$. Both terms are now negative, so they accumulate in the same direction: the distances add, $3.5 + 2.25 = 5.75$, and the common sign is negative. -5.75

Watch for: an answer of -1.25 means the subtraction was rewritten as an addition without changing the sign of the second number.

4. Compare $-8 - (-3)$ and $-8 + 3$.

Solution: Rewrite the left side by adding the opposite: $-8 - (-3) = -8 + 3$. The two expressions are now identical character for character, so no arithmetic is needed — they must be equal. (Both do evaluate to -5 .) The symbol that belongs in the blank is $\boxed{=}$.
Completed sentence: subtracting -3 has the same effect as *adding* 3, because the opposite of -3 is 3.

5. Order $-2 - 7$, $2 - 7$, $-2 + 7$ from least to greatest.

Solution: Evaluate each one first, adding the opposite where needed. $-2 - 7 = -2 + (-7) = -9$. $2 - 7 = 2 + (-7) = -5$. $-2 + 7 = 5$. On the number line -9 sits farthest left, then -5 , then 5 to the right of zero. $\boxed{-9 < -5 < 5}$

Watch for: ordering by digits alone puts -9 last. Among negatives, the larger the digit, the smaller the number.

6. Find and fix: Dana wrote $-4 - (-9) = -13$.

Solution: Dana added the two negatives instead of subtracting a negative — she computed $-4 + (-9)$. The rule she broke is “subtract by adding the opposite”: the opposite of -9 is $+9$, not -9 . Corrected, $-4 - (-9) = -4 + 9$, and moving 9 units right from -4 lands 5 units past zero. $\boxed{5}$

A useful size check: subtracting a negative always makes the result *larger* than the number you started with, so any answer below -4 is wrong before the arithmetic is even checked.

7. Find and fix: Marcus wrote $-7 - 4 = -3$.

Solution: Marcus treated $-7 - 4$ as $-7 + 4$; he used the minus as a sign on the 4 and then added it anyway. Subtracting a positive means adding a negative: $-7 - 4 = -7 + (-4)$. Both terms are negative, so the distances add, $7 + 4 = 11$, with a negative sign. $\boxed{-11}$

Size check: subtracting a positive must move the result *left*, so the answer has to be less than -7 ; -3 is to the right of -7 and cannot be correct.

8. Challenge — find and fix: a student got -12.75 for $-2.5 - (-6) - 4.25$.

Solution: The mishandled sign is the one in the middle term. The student read $-(-6)$ as -6 , computing $-2.5 - 6 - 4.25 = -12.75$; the double negative should have become $+6$. Rewrite every subtraction as an addition first, then work left to right:

$$-2.5 - (-6) - 4.25 = -2.5 + 6 + (-4.25)$$

$$-2.5 + 6 = 3.5$$

$$3.5 + (-4.25) = -0.75$$

The correct value is

$$\boxed{-0.75}$$

The corrected answer is 12 units to the right of the student's, which is exactly twice the 6 whose sign was flipped — the standard signature of a single dropped double negative.